

Auto Ticket Classification Using Flow Designer

Project Description:

Auto Ticket Classification Using Flow Designer is a ServiceNow-based automation project that classifies incoming support tickets automatically and routes them to the appropriate category, assignment group, priority, and workflow. The solution uses Flow Designer to reduce manual ticket handling, improve consistency, and speed up incident processing.

The flow can evaluate ticket information such as short description, description, category, service, impact, urgency, and keywords. Based on defined conditions, it updates classification fields and assigns the ticket to the responsible support team.

Scenarios

Scenario 1: A user submits a ticket describing that a laptop cannot connect to the company network. The flow identifies the request as a network-related incident and routes it to the appropriate network support group.

Scenario 2: A ticket reports that an application is not opening. Flow Designer evaluates the ticket details, classifies it as an application issue, sets the relevant priority according to the defined conditions, and assigns it to the application support team.

Scenario 3: A password-related access ticket is created. The flow detects the relevant keywords/category, classifies the ticket as an access or authentication issue, and routes it to the service desk or identity support group.

Technical Architecture

The solution uses ServiceNow Flow Designer as the automation layer. A ticket is created or updated in the Incident table, which triggers the flow. The flow reads ticket fields, evaluates conditions, determines the classification, updates the record, and assigns the ticket to the appropriate support group.

- ServiceNow Incident table – stores incoming support tickets.
- Flow Designer – provides the trigger, conditions, actions, and automation logic.
- Classification rules – map ticket information to categories and support groups.
- Assignment logic – routes tickets to the appropriate team.
- Notifications – optionally inform users or support teams after classification.
- Audit/history – provides traceability of automated updates.

Pre-requisites:

- Basic knowledge of ServiceNow Incident Management.
- Understanding of Flow Designer and flow triggers.
- Knowledge of tables, fields, conditions, and assignment groups.
- Access to a ServiceNow instance with permission to create and test flows.
- Understanding of incident categories, priorities, impact, and urgency.
- Basic testing and troubleshooting knowledge.

Project Workflow

Milestone 1: Requirement Analysis and Design

- Activity 1.1: Identify the ticket types that need automatic classification.
- Activity 1.2: Define classification criteria such as keywords, category, service, impact, and urgency.
- Activity 1.3: Map each ticket type to the appropriate assignment group.
- Activity 1.4: Define the expected priority and classification outcomes.

Milestone 2: Flow Designer Configuration

- Activity 2.1: Create a new flow in Flow Designer.
- Activity 2.2: Configure the Incident-created or Incident-updated trigger.
- Activity 2.3: Add conditions to identify ticket types.
- Activity 2.4: Add actions to update classification and assignment fields.

Milestone 3: Classification Logic Development

- Activity 3.1: Configure condition branches for network, application, access, hardware, and other ticket types.
- Activity 3.2: Set category/subcategory values.
- Activity 3.3: Set assignment group based on classification.
- Activity 3.4: Configure priority using impact and urgency where required.

Milestone 4: Testing

- Activity 4.1: Create sample tickets for each classification.
- Activity 4.2: Verify that each ticket follows the correct flow path.
- Activity 4.3: Check category, priority, assignment group, and other updated fields.
- Activity 4.4: Test exception and unmatched-ticket scenarios.

Milestone 5: Deployment and Monitoring

- Activity 5.1: Activate the tested flow.
- Activity 5.2: Monitor flow executions and failed actions.
- Activity 5.3: Review classification accuracy and update rules when necessary.
- Activity 5.4: Document the final flow and classification rules.

Milestone 1: Requirement Analysis and Design

The first milestone establishes what the automation must classify and how each ticket should be handled. The classification design should be simple, consistent, and easy to maintain.

Activity 1.1: Identify Ticket Types

Review common incidents and group them into meaningful classifications such as Hardware, Network, Application, Access, Email, and Other.

Activity 1.2: Define Classification Criteria

For each classification, identify reliable fields and keywords. For example, terms such as Wi-Fi, network, VPN, or connection can be used as indicators for network-related tickets.

Activity 1.3: Map Assignment Groups

Create a mapping between each classification and the responsible assignment group. This allows the flow to route tickets without manual intervention.

Activity 1.4: Define Priority Rules

Define how impact and urgency should influence priority. The exact priority matrix should follow the organization's ServiceNow configuration and business rules.

Milestone 2: Flow Designer Configuration

Activity 2.1: Create the Flow

Open Flow Designer and create a new flow with a clear name such as “Auto Ticket Classification”. Add a description explaining that the flow classifies and routes incident tickets automatically.

Activity 2.2: Configure the Trigger

Select the Incident table and configure the trigger for when an incident is created. If required, the flow can also run when an incident is updated, subject to safeguards that prevent unnecessary repeated execution.

Activity 2.3: Read Ticket Information

Use the trigger record as the input for classification. Relevant data can include Short description, Description, Category, Subcategory, Impact, Urgency, Service, and Caller information.

Activity 2.4: Add Classification Conditions

Use Flow Designer condition logic to evaluate the ticket. Create branches for the defined classifications and ensure that the most specific conditions are evaluated appropriately.

Milestone 3: Classification Logic Development

Example Classification Logic

- **Network:** ticket information contains network, Wi-Fi, VPN, connection, or similar indicators.
- **Hardware:** ticket indicates laptop, desktop, monitor, keyboard, mouse, printer, or device failure.
- **Application:** ticket indicates application error, software failure, or application access problem.
- **Access:** ticket indicates password, login, account, permission, or authentication issue.
- **Other:** used when the ticket does not satisfy the defined classification conditions.

Flow Actions

- Update the Incident Category and Subcategory.
- Set or adjust Priority according to approved rules.
- Set Assignment Group based on the ticket classification.
- Optionally add a work note explaining the automated classification.
- Optionally send a notification to the assigned support group.

The flow should be designed so that classification and routing are transparent and traceable. Where multiple conditions can match, define an explicit order or mutually exclusive conditions to avoid conflicting updates.

Milestone 4: Testing and Validation

Activity 4.1: Prepare Test Cases

- Create a network incident and verify network classification.
- Create a hardware incident and verify hardware classification.
- Create an application incident and verify application classification.
- Create an access incident and verify access classification.
- Create an unmatched incident and verify the default handling.

Activity 4.2: Validate Flow Execution

Use Flow Designer execution details to verify which trigger, conditions, and actions were executed. Confirm that the expected branch was selected and that record updates were completed successfully.

Activity 4.3: Exception Testing

Test incomplete descriptions, unexpected wording, missing fields, and tickets that could match multiple categories. Review failed executions and refine conditions where necessary.

Milestone 5: Deployment and Monitoring

After successful testing, activate the flow and monitor its executions. Automated classification should be reviewed periodically to ensure that the rules continue to reflect actual ticket patterns and organizational assignment groups.

Deployment Checklist

- All conditions tested.
- Assignment groups verified.
- Priority rules verified.
- Default/unmatched handling configured.
- Flow execution errors reviewed.
- Flow activated only after testing.
- Documentation completed.

Monitoring

Monitor flow execution history for failures, unexpected branches, and repeated executions. Use real ticket examples to identify missing keywords or classification rules and improve the flow over time.

Exploring the Automated Ticket Classification Flow

Input: An incident is created with ticket information such as short description and description.

Trigger: Flow Designer starts the flow when the configured incident trigger condition is met.

Decision: The flow evaluates classification conditions.

Action: The selected branch updates category, subcategory, priority, and assignment group according to the configured rules.

Output: The incident is automatically classified and routed to the appropriate support team.

Example: If a user creates an incident with a description such as "VPN is not connecting", the flow can identify the network/access-related classification according to the configured conditions and route it to the responsible group.

Conclusion

Auto Ticket Classification Using Flow Designer provides a structured approach to automating incident classification and routing in ServiceNow. By combining triggers, conditions, and actions, the solution can reduce repetitive manual work and create a consistent ticket-handling process.

The project covers requirement analysis, Flow Designer configuration, classification logic, testing, deployment, and monitoring. The approach can be extended with additional classification rules, reusable subflows, notifications, approval logic, reporting, and more advanced classification mechanisms as the organization's requirements evolve.

Overall, the project demonstrates how Flow Designer can be used to automate a practical IT service-management process while maintaining clear, configurable, and traceable workflow logic.